

$$\begin{array}{c}
\mathcal{K}_{0^+}^{\#1} \quad \mathcal{K}_{0^+}^{\#2} \\
\mathcal{K}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \frac{k^2 \binom{(4)}{\kappa_3}}{3} & -\frac{k^2 \binom{(4)}{\kappa_3}}{3} \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline -\frac{k^2 \binom{(4)}{\kappa_3}}{3} & \frac{k^2 \binom{(4)}{\kappa_3}}{3} \\ \hline \end{array} \quad \mathcal{K}_{1^+}^{\#1}{}_\alpha \quad \mathcal{K}_{1^+}^{\#2}{}_\alpha \\
\mathcal{K}_{1^+}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline -\frac{\Upsilon_1 k^2}{9} & -\frac{1}{9} \sqrt{5} \Upsilon_2 k^2 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline -\frac{1}{9} \sqrt{5} \Upsilon_2 k^2 & -\frac{5 k^2 \binom{(4)}{\kappa_1}}{3} \\ \hline \end{array} \quad \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} \\
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|} \hline \frac{k^2 \binom{(4)}{\kappa_3}}{3} \\ \hline \end{array} \quad \mathcal{K}_{3^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{3^+}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline 0 \\ \hline \end{array}
\end{array}$$

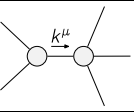
$$\begin{array}{c}
\mathcal{J}_{0^+}^{\#1} \quad \mathcal{J}_{0^+}^{\#2} \\
\mathcal{J}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \frac{3}{4 k^2 \binom{(4)}{\kappa_3}} & -\frac{3}{4 k^2 \binom{(4)}{\kappa_3}} \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline -\frac{3}{4 k^2 \binom{(4)}{\kappa_3}} & \frac{3}{4 k^2 \binom{(4)}{\kappa_3}} \\ \hline \end{array} \quad \mathcal{J}_{1^+}^{\#1}{}_\alpha \quad \mathcal{J}_{1^+}^{\#2}{}_\alpha \\
\mathcal{J}_{1^+}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline -\frac{5 k^2 \binom{(4)}{\kappa_1}}{3 \text{Det}(1^+)} & \frac{\sqrt{5} \Upsilon_2 k^2}{9 \text{Det}(1^+)} \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline \frac{\sqrt{5} \Upsilon_2 k^2}{9 \text{Det}(1^+)} & -\frac{\Upsilon_1 k^2}{9 \text{Det}(1^+)} \\ \hline \end{array} \quad \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta} \\
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|} \hline \frac{3}{k^2 \binom{(4)}{\kappa_3}} \\ \hline \end{array} \quad \mathcal{J}_{3^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{3^+}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline 0 \\ \hline \end{array}
\end{array}$$

Abbreviations used in matrices

$$\Upsilon_1 == 3 \binom{(4)}{\kappa_1} - 4 \binom{(4)}{\kappa_3} \ \&\& \ \Upsilon_2 == 3 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_3} \ \&\& \ \text{Det}(1^+) == -\frac{5}{81} k^4 \binom{(4)}{\kappa_3} (18 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_3})$$

Lagrangian

$$\binom{(4)}{\kappa_1} \partial_\beta \mathcal{K}_\chi{}^\delta \partial^\chi \mathcal{K}_\alpha{}^\beta - \binom{(4)}{\kappa_1} \partial_\chi \mathcal{K}_\beta{}^\delta \partial^\chi \mathcal{K}_\alpha{}^\beta + \binom{(4)}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta - \frac{2}{3} \binom{(4)}{\kappa_3} \partial^\chi \mathcal{K}_\alpha{}^\beta \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#2} + 3 \mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta \mathcal{J}_\alpha{}^\beta == 0$	
$\mathcal{J}_{3^+}^{\#1\alpha\beta\chi} == 0$	7	$ \begin{aligned} & k^\delta k^\epsilon (3 \partial_\epsilon \partial^\chi \partial^\beta \partial^\alpha \mathcal{J}_\delta{}^\phi - \partial_\epsilon \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^{\chi\phi}{}_\phi - \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\beta\phi}{}_\phi - \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\alpha\phi}{}_\phi + 2 \partial_\phi \partial^\chi \partial^\beta \partial^\alpha \mathcal{J}_{\delta\epsilon}{}^\phi - \\ & 4 \partial_\phi \partial_\epsilon \partial^\beta \partial^\alpha \mathcal{J}^{\chi}{}_\delta{}^\phi - 4 \partial_\phi \partial_\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\beta}{}_\delta{}^\phi - 4 \partial_\phi \partial_\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\alpha}{}_\delta{}^\phi + 5 \partial_\phi \partial_\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi}{}_\phi + 5 \partial_\phi \partial_\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi}{}_\phi + \\ & 5 \partial_\phi \partial_\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\phi} - 5 \partial_\phi \partial^\phi \partial_\delta \mathcal{J}^{\alpha\beta\chi} - \eta^{\beta\chi} \partial_\nu \partial_\epsilon \partial_\delta \partial^\alpha \mathcal{J}^\phi{}_\nu{}^\gamma - \eta^{\alpha\chi} \partial_\nu \partial_\epsilon \partial_\delta \partial^\beta \mathcal{J}^\phi{}_\nu{}^\gamma - \eta^{\alpha\beta} \partial_\nu \partial_\epsilon \partial_\delta \partial^\chi \mathcal{J}^\phi{}_\nu{}^\gamma + \\ & \eta^{\beta\chi} \partial_\nu \partial_\phi \partial_\epsilon \partial^\alpha \mathcal{J}_\delta{}^\phi{}_\nu + \eta^{\alpha\chi} \partial_\nu \partial_\phi \partial_\epsilon \partial^\beta \mathcal{J}_\delta{}^\phi{}_\nu + \eta^{\alpha\beta} \partial_\nu \partial_\phi \partial_\epsilon \partial^\chi \mathcal{J}_\delta{}^\phi{}_\nu - \eta^{\beta\chi} \partial_\nu \partial_\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\alpha\phi}{}_\nu - \eta^{\alpha\chi} \partial_\nu \partial_\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\beta\phi}{}_\nu - \\ & \eta^{\alpha\beta} \partial_\nu \partial_\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\chi\phi}{}_\nu + \eta^{\beta\chi} \partial_\nu \partial^\nu \partial_\epsilon \partial_\delta \mathcal{J}^{\alpha\phi}{}_\phi + \eta^{\alpha\chi} \partial_\nu \partial^\nu \partial_\epsilon \partial_\delta \mathcal{J}^{\beta\phi}{}_\phi + \eta^{\alpha\beta} \partial_\nu \partial^\nu \partial_\epsilon \partial_\delta \mathcal{J}^{\chi\phi}{}_\phi) == 0 \end{aligned} $	
Total # constraint(s):	8		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{1}{18 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_3}}$
Resolved unitarity condition(s):		$\binom{(4)}{\kappa_1} > -\frac{\binom{(4)}{\kappa_3}}{18}$	